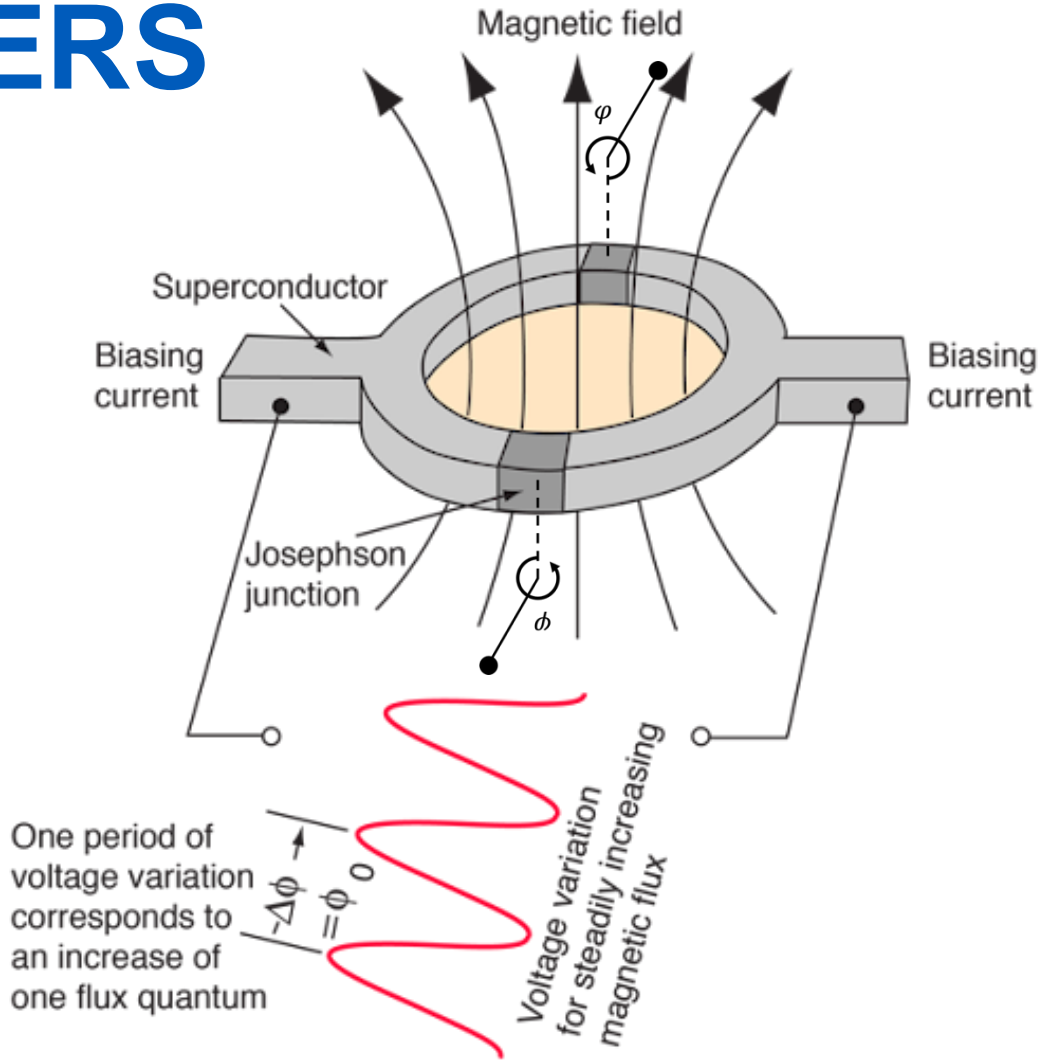
SQUID MAGNETOMETERS

A WAY TO MEASURE THE BRAIN

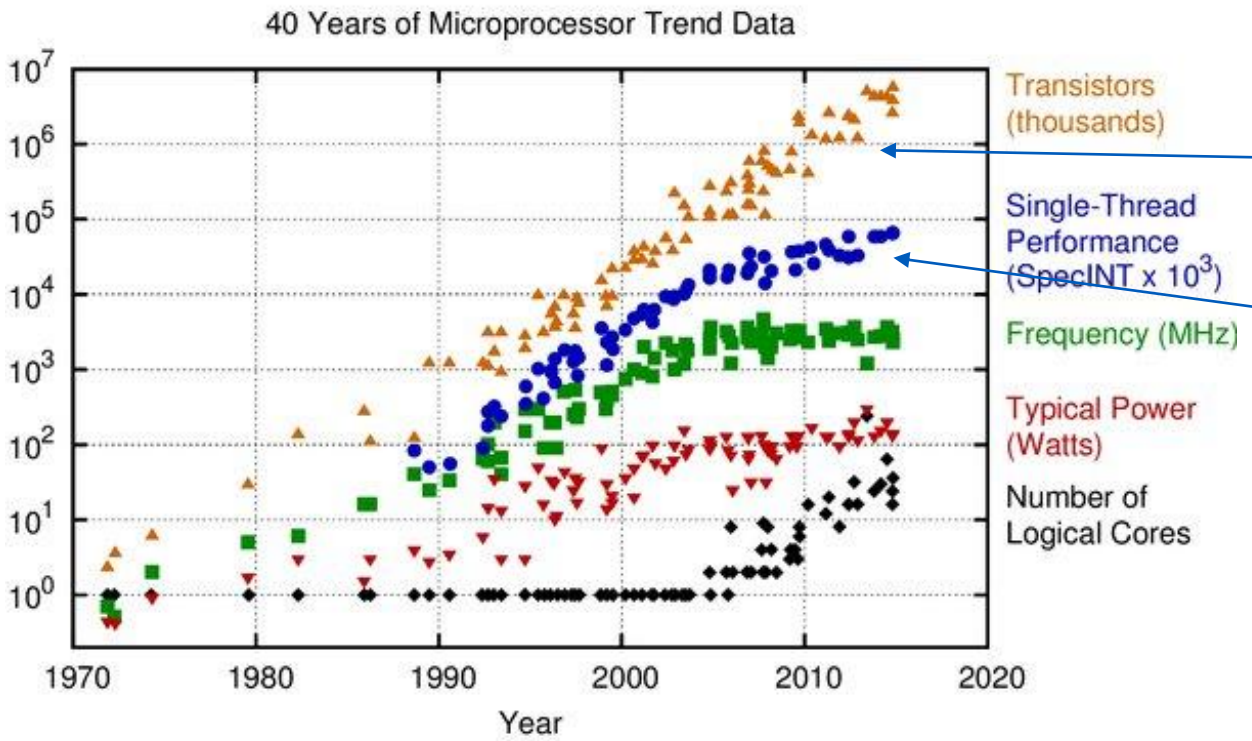
Dario Tringali



Overview

- Emergence of Superconducting Devices
- What are Josephson Junctions? How do they work?
- Creating a Model
- Magnetometers: SQUIDs
- SIMULATIONS!!

Moore's Law



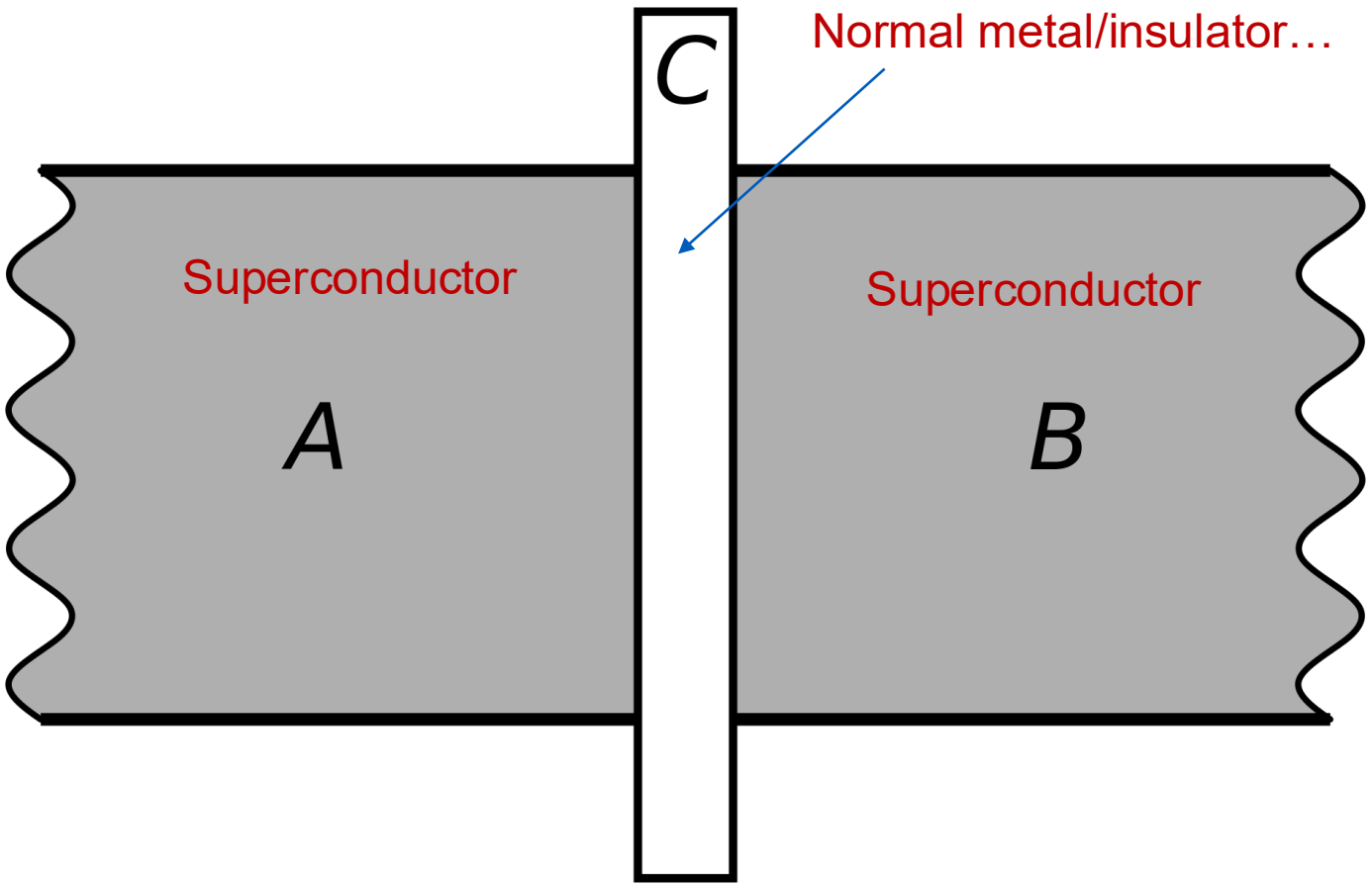
Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2015 by K. Rupp

Linear trend

Plateau

Performance gain starting to level off

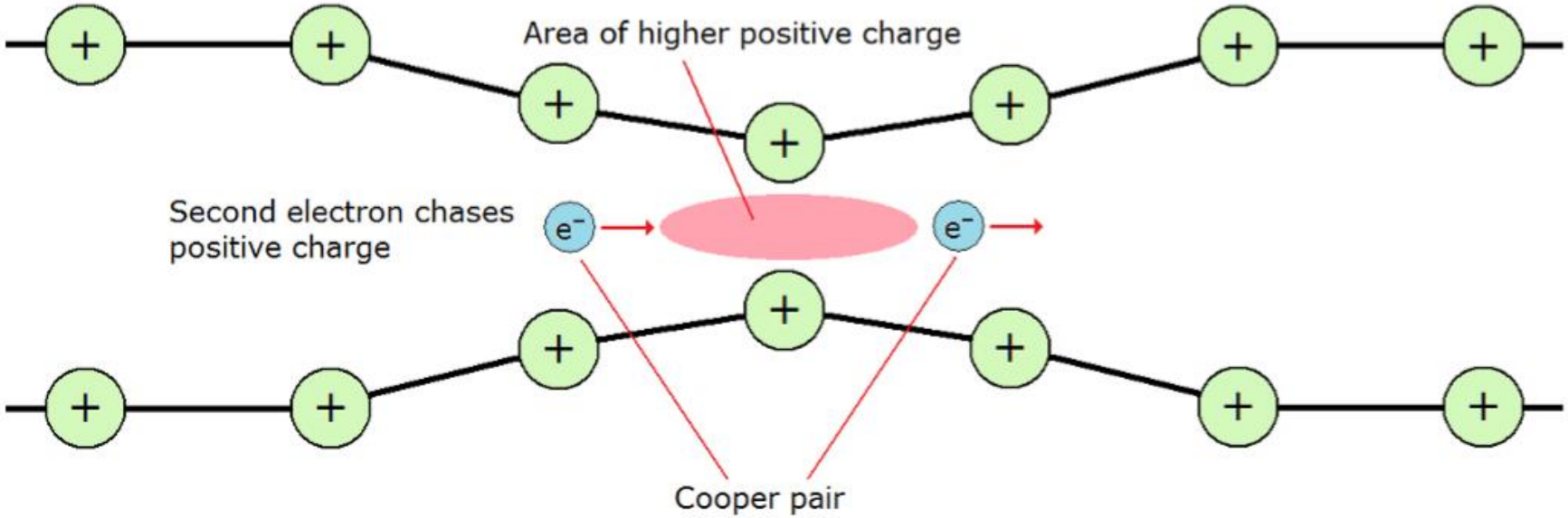
The Josephson Junction (JJ)



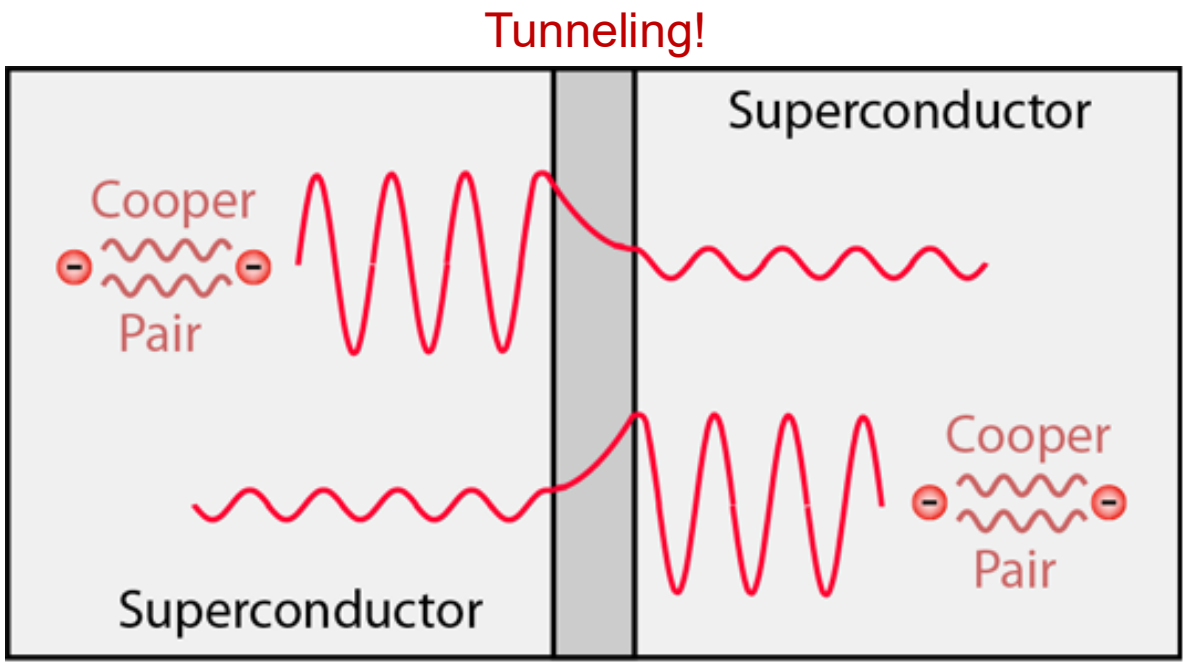
Cooper Pairs

- Condensation of Cooper Pairs lead to super conductivity

- Phonon-Electron interaction



The Josephson Effect

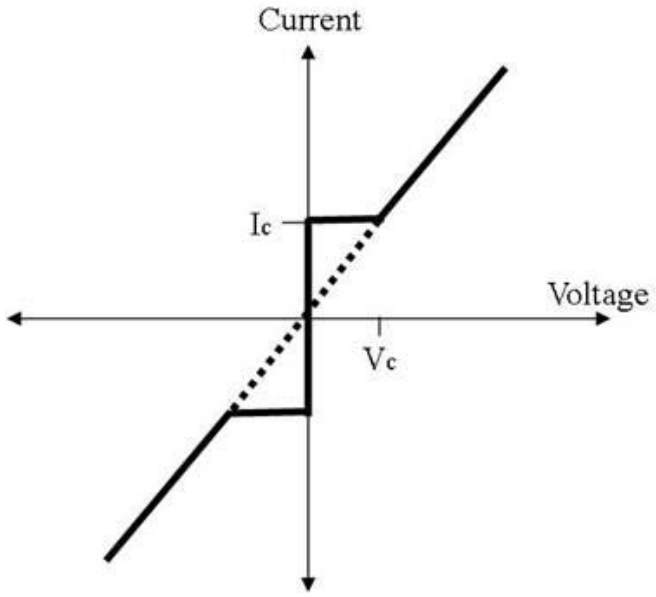


Super Current: $I_s = I_C \sin(\varphi)$

- φ is the relative phase between the two superconducting regions
- Present with no applied voltage

Adding a Voltage

- Phase can be changed according to: $\frac{d\varphi}{dt} = \frac{2\pi e}{\hbar} V = \frac{2\pi}{\Phi_0} V$
- Magnetic Flux Quantum: $\Phi_0 = \frac{h}{2e}$
- Application of voltage leads to normal current: $I_N = \frac{V}{R_N}$



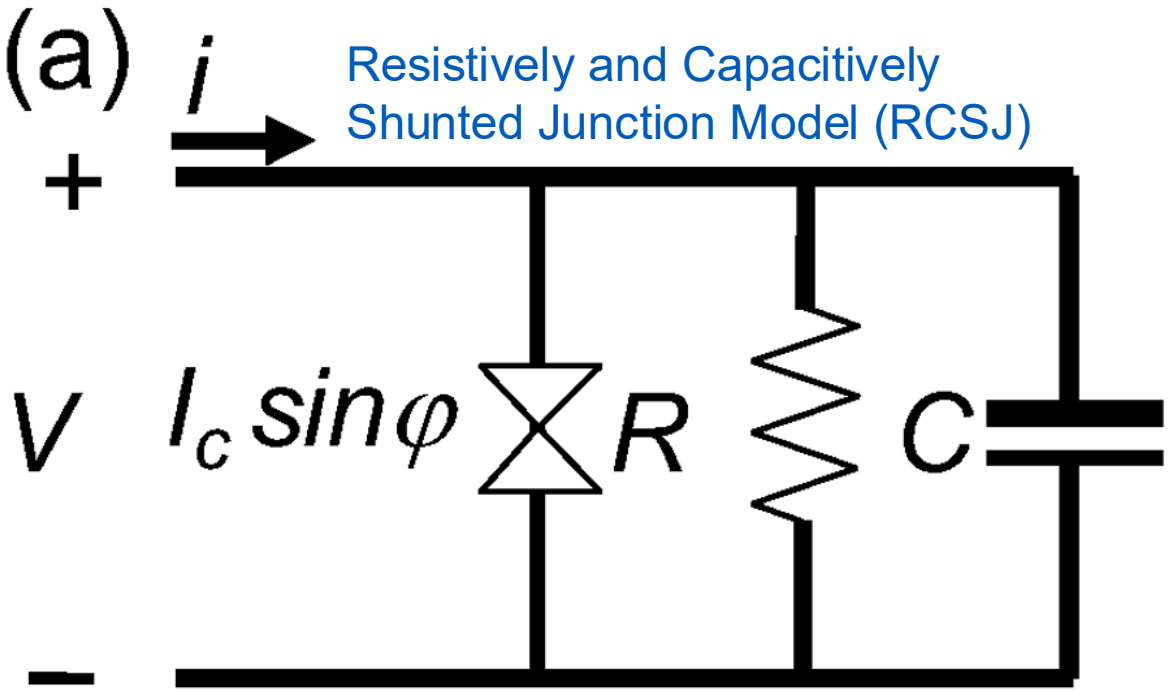
Changing the Voltage

- Generates a displacement current: $I_D = C\frac{dV}{dt}$

- Determined by geometry of the non-superconducting bridge and its electric permittivity

Let's combine all these currents together…

RCSJ Model



Resistively and Capacitively Shunted Junction Model (RCSJ)

$$i = \frac{I}{I_C} = I_s + I_N + I_D = I_C \sin(\varphi) + \frac{V}{R_N} + C\frac{dV}{dt}$$

If we "massage" this model, we can relate the current i to the phase φ (CPR)

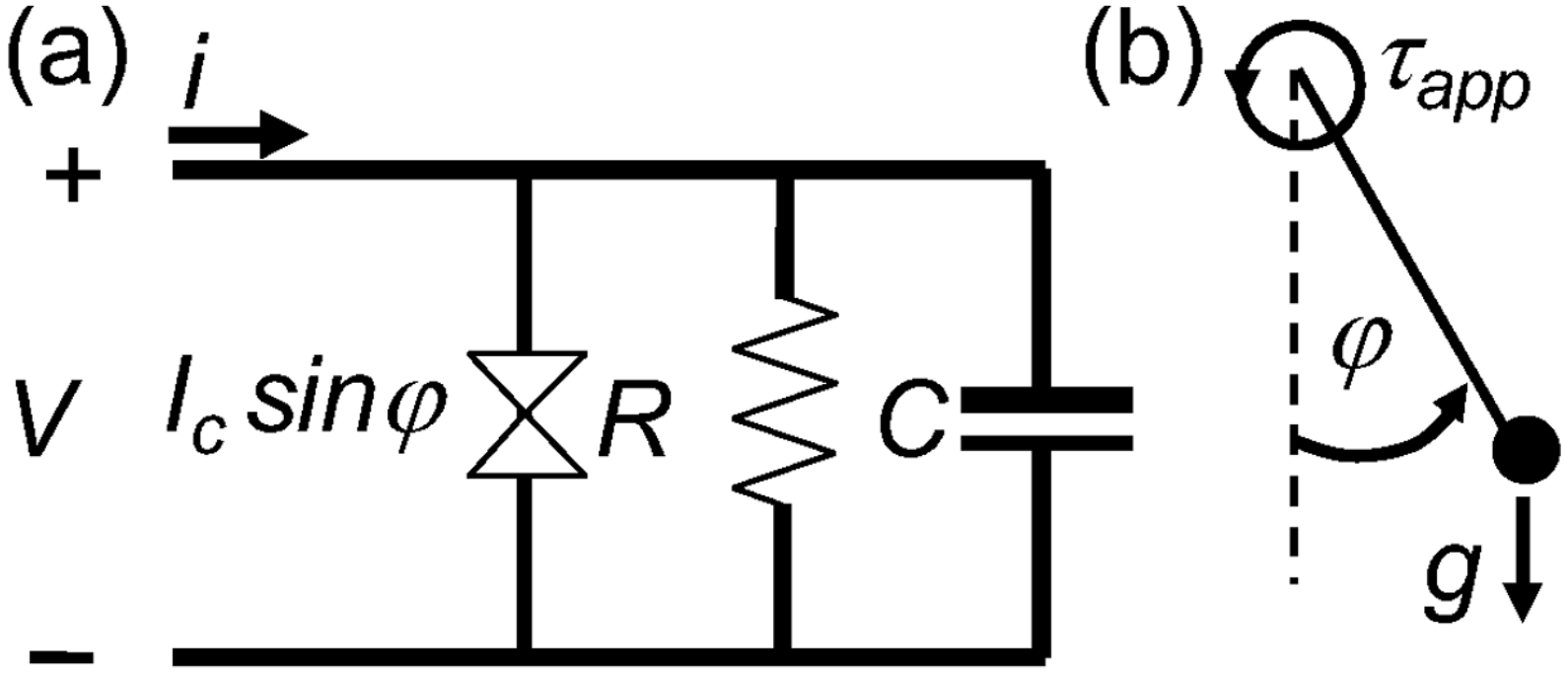
$$i = \sin(\varphi) + \alpha\frac{d\varphi}{d\tau} + \frac{d^2\varphi}{d\tau^2}$$

where have we seen this equation before…

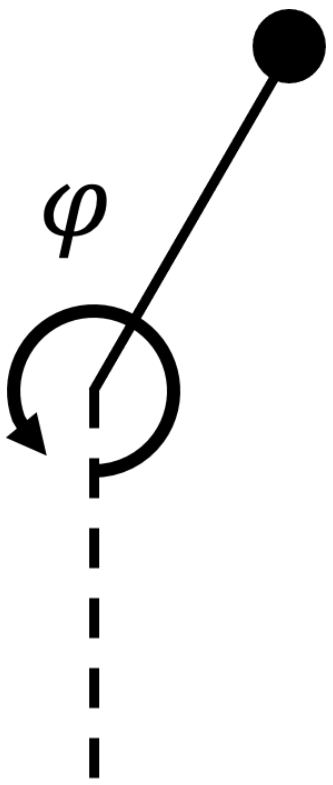
Pendulum Analogy

$$i = \sin(\varphi) + \alpha \frac{d\varphi}{d\tau} + \frac{d^2\varphi}{d\tau^2}$$

Analogous to a driven damped pendulum!!



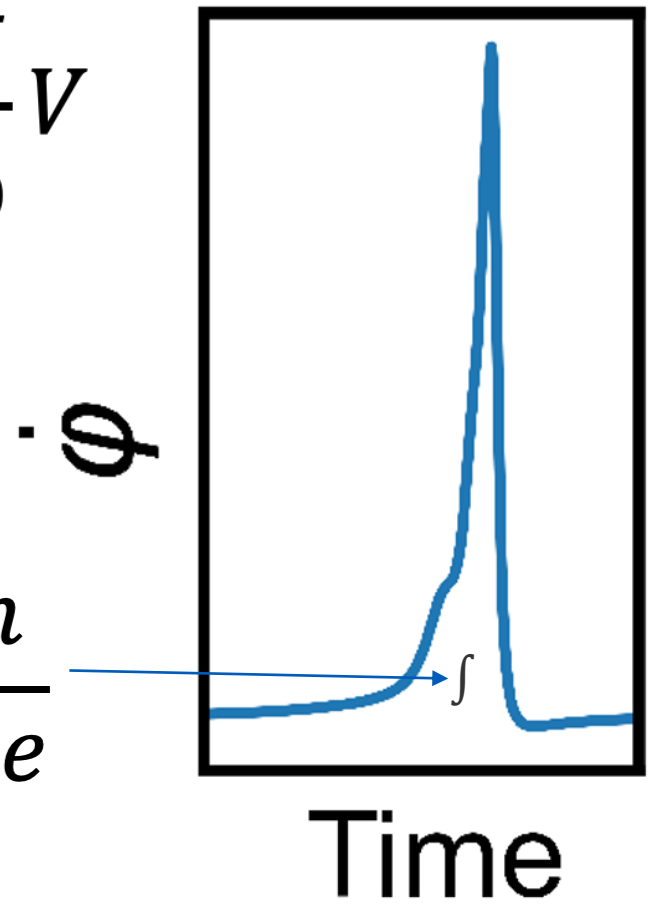
SFQ Pulses



- Making a full 2π rotation of pendulum creates a voltage spike
- Can make superconducting digital circuits
- Can perform switching 1000x faster than transistors

$$\dot{\varphi} = \frac{2\pi}{\Phi_0} V$$

$$\Phi_0 = \frac{h}{2e}$$



i_b: The total external bias current
Q: The quality factor of the JJ

Simulating a Single JJ

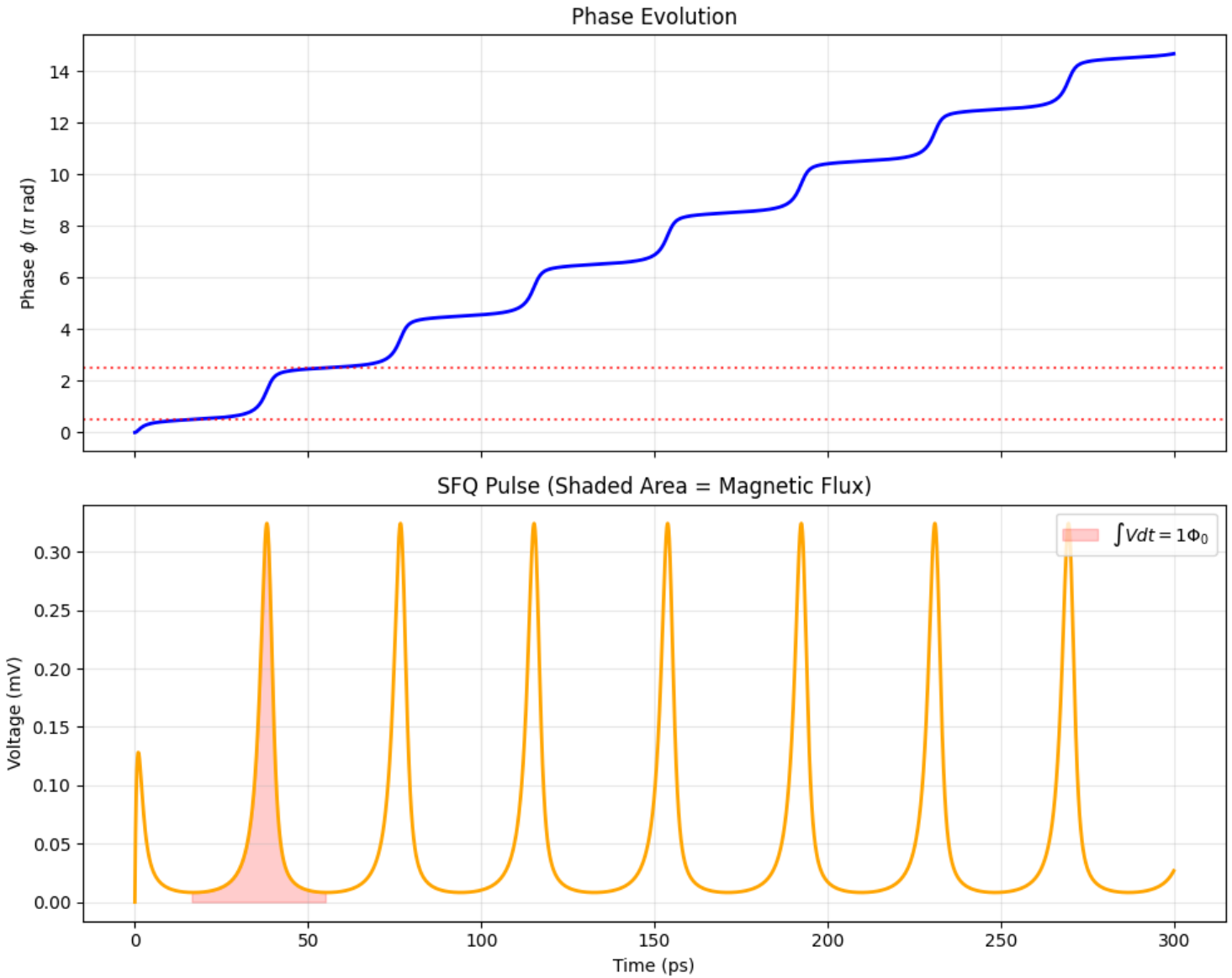
- Can simulate using just RK4!!
- Put diff eq in terms of dimensionless parameters:

$$\ddot{\varphi} = i_b - \frac{1}{Q}\dot{\varphi} - \sin\varphi$$

1. **Initialize State:** Define the starting phase (φ) and velocity $(\dot{\varphi})$ at time zero.
2. **Calculate Acceleration:** Solve the RCSJ differential equation to find the instantaneous phase acceleration $(\ddot{\varphi})$
3. **Euler Update:** Multiply $\ddot{\varphi}$ by the time step $(d\tau)$ to push $\dot{\varphi}$ and φ forward into the future
4. **Loop & Convert:** Repeat steps 2 and 3 for the entire time window, then convert the final dimensionless velocity into physical voltage

$$\dot{\varphi} = \frac{2\pi}{\Phi_0} V$$

Simulating a Single Josephson Junction



Code Output:

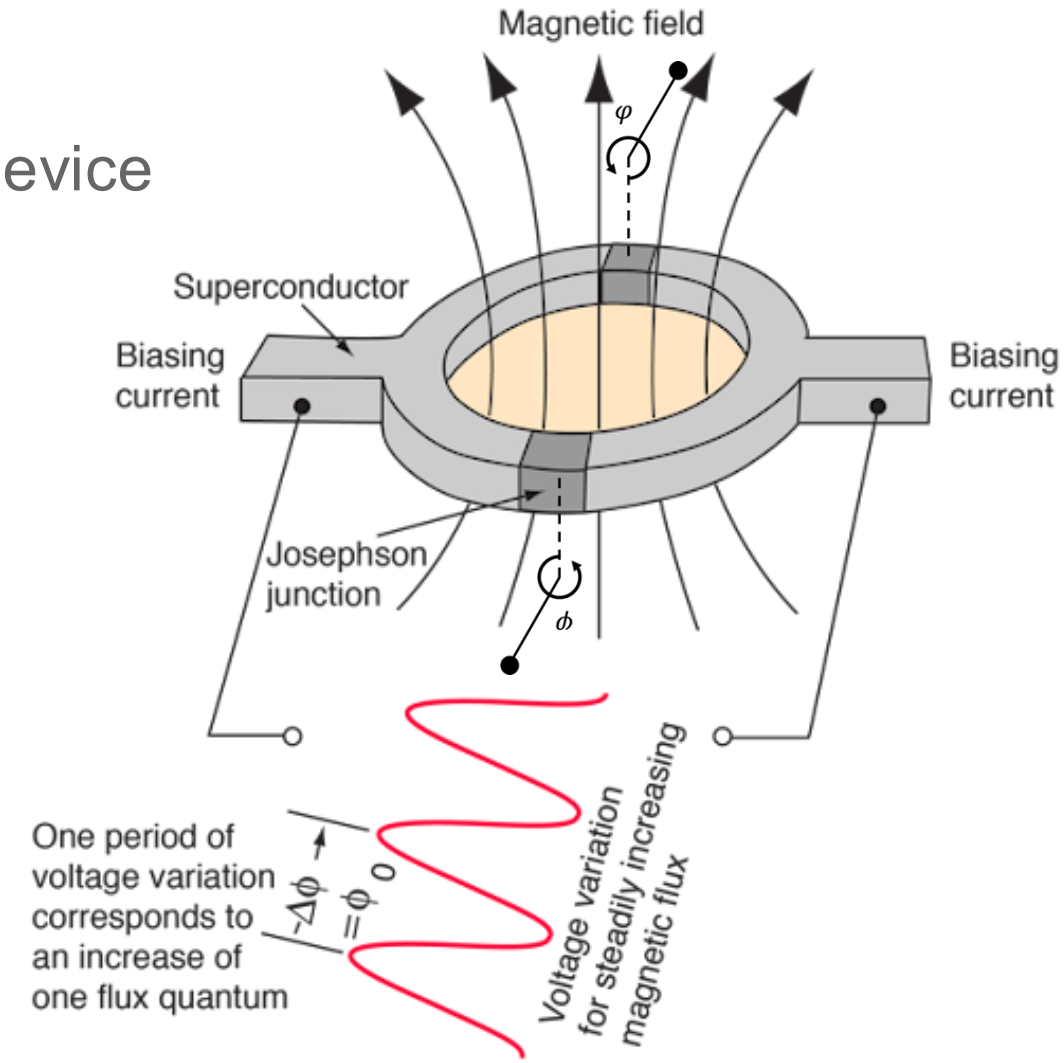
Integrating across 1 SFQ Pulse...
Calculated Φ_0 : 2.06762e-15 Wb
Theoretical Φ_0: 2.06783e-15 Wb
Error : 0.01012 %

SQUIDs

Superconducting Quantum Interference Device

- Threshold for SQUID: $10^{-14}\ T$
- Magnetic field of heart: $10^{-10}\ T$
- Magnetic field of brain: $10^{-13}\ T$

$$\Phi_0 = 2.0678 \cdot 10^{-15}\ T \cdot m^2$$



Simulating SQUIDs

1. **Initialize Dual States:** Track phase (φ) and velocity ($\dot{\varphi}$) for *two* separate junctions simultaneously.
2. **Calculate Coupling:** Compute the circulating current (j_{circ}) driven by the external magnetic flux.
3. **Split Effective Current:** Divide the total bias current, modifying each junction by the circulating current.
4. **Synchronous Euler Update:** Solve the RCSJ acceleration ($\ddot{\varphi}$) for both junctions and step them forward in time together.
5. **Average the Output:** Average the two individual junction voltages to find the macroscopic SQUID voltage.

$\boldsymbol{\Phi_{ext}}$: External magnetic flux (normalized to Φ_0)
$\boldsymbol{\beta_L}$: Geometric inductance of the SQUID loop
$\boldsymbol{i_b}$: The total external bias current
$\boldsymbol{j_{circ}}$: Circulating current between the JJs
$\boldsymbol{i_{eff}}$: Net current experienced by a JJ

1. Circulating Current:

$$j_{circ} = \frac{\phi_1 - \phi_2 - 2\pi\Phi_{ext}}{\pi\beta_L}$$

2. Effective Junction Currents:

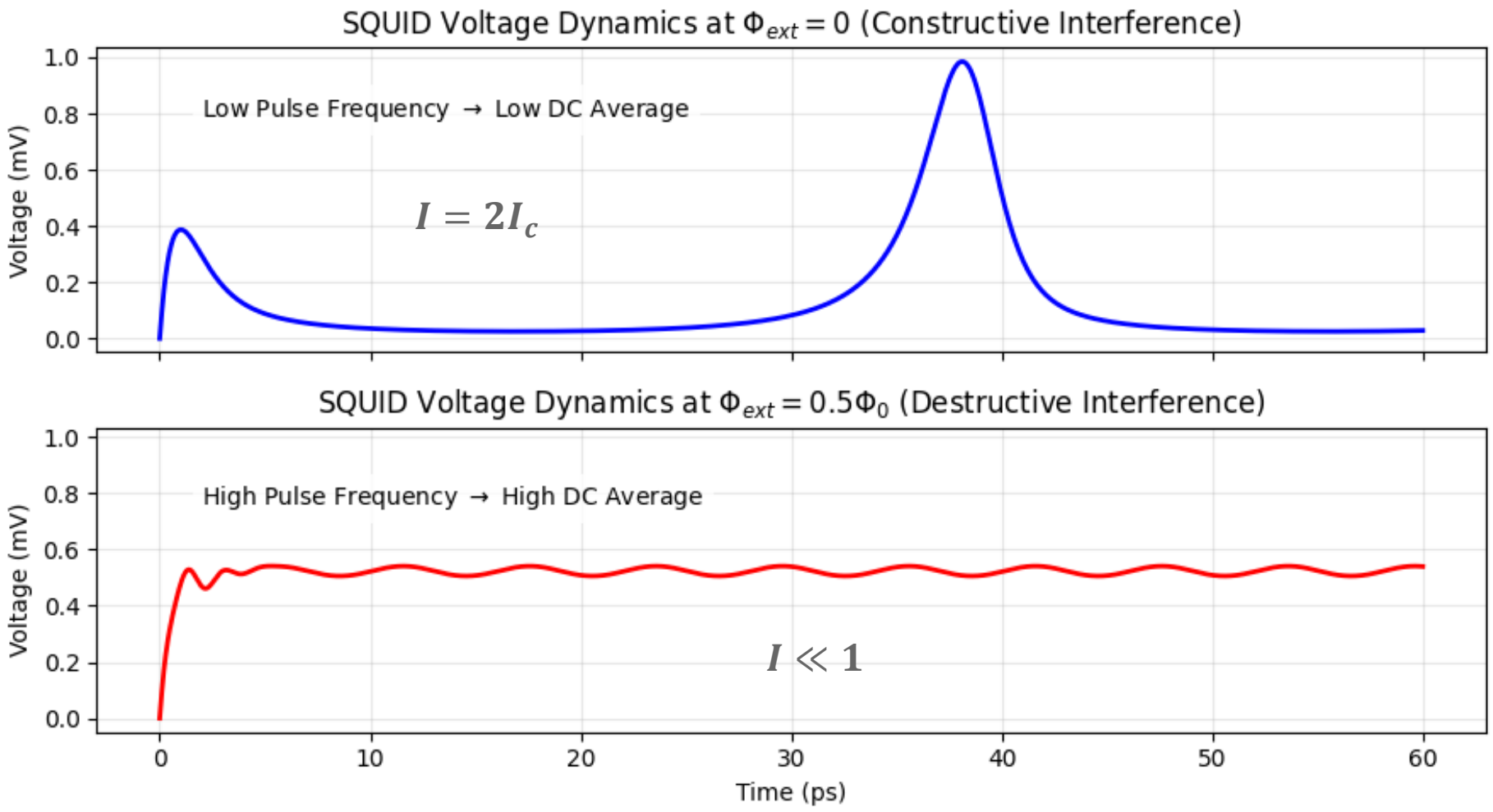
$$i_{eff,1} = \frac{i_b}{2} - j_{circ}$$

3. Simulate Each JJ with Effective Current:

$$\ddot{\varphi} = i_{eff} - \frac{1}{Q}\dot{\varphi} - \sin\varphi$$

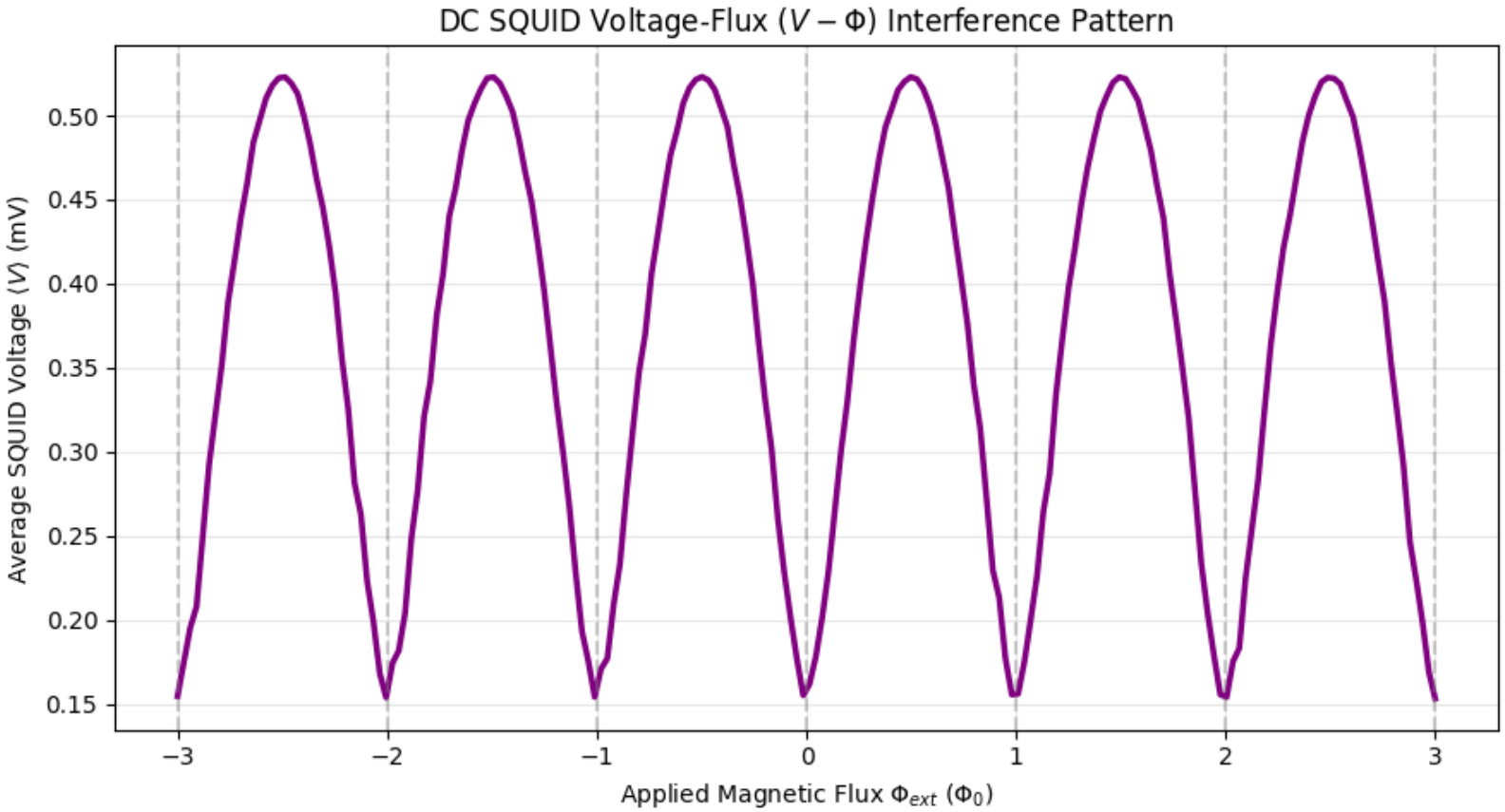
Simulating SQUIDs

$i_b = 2.1 I_c$



Simulating SQUIDs

- The SQUID's oscillating response creates ambiguity when measuring absolute magnetic flux.
- We cannot simply count voltage peaks because real signals are microscopic fractions of a single flux quantum.
- How can we accurately measure the exact external magnetic field going through the SQUID?



Feedback Loop!!

Measuring Absolute Field: Flux-Locked Loop (FLL)

1. **Read Voltage:** Average the high-frequency quantum dynamics to find the macroscopic DC voltage ($\langle V \rangle$).
2. **Calculate Error:** Compare the measured voltage against a highly sensitive setpoint $(V_{setpoint})$ on the steepest part of the $V - \Phi$ curve.
3. **Drive Current:** The PI controller pushes a physical current (I_{fb}) to minimize the error.
4. **Cancel Flux:** A feedback coil with mutual inductance (M_f) generates an opposing magnetic field (Φ_{fb}) to hold the SQUID exactly at the setpoint.
5. **Measure the Field:** Because the loop forces Φ_{net} to be constant, any change in external flux is perfectly mirrored by the feedback flux. Therefore: $\boldsymbol{\Delta\Phi_{ext} = M_f \Delta I_{fb}}$

1. Net Flux (The SQUID Input)

$$\Phi_{net} = \Phi_{ext} - \Phi_{fb}$$

2. Error Signal

$$e(t) = \langle V(\Phi_{net}) \rangle - V_{setpoint}$$

3. PI Controller (The Op-Amp)

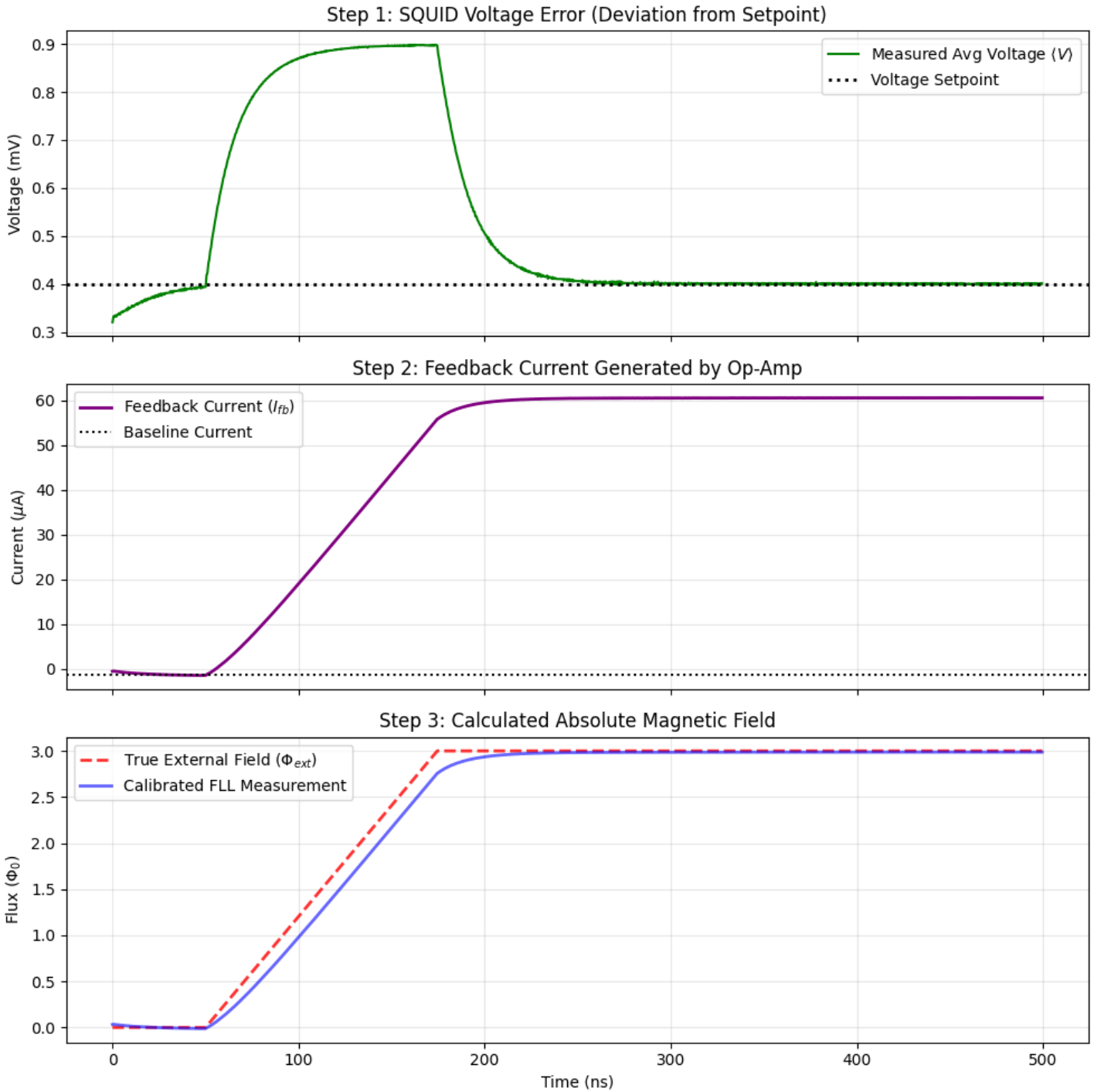
$$I_{fb} = K_p e(t) + K_i \int e(t) dt$$

4. Feedback Coupling

$$\Delta\Phi_{ext} = \Phi_{fb} = M_f I_{fb}$$

FLL Loop Sim

- The PI controller effectively zeroes out the error, generating the exact feedback current needed to hold the SQUID at the setpoint
- It tracks a $3\Phi_0$ field ramp, this can bypass the quantum phase ambiguity of the $V - \Phi$ curve.
- By measuring the linear feedback current instead of the SQUID's oscillating voltage, we extracted the external magnetic field!!



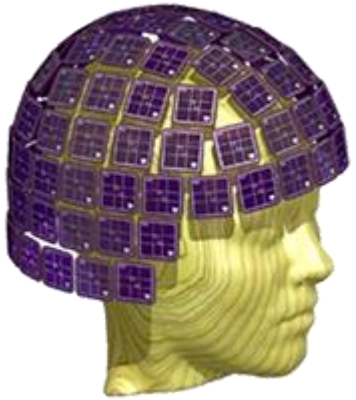
Magnetoencephalography (MEG)

- Use an array of SQUIDs to map brain activity

- Gives good temporal resolution due to the high acquisition rate of SQUIDs

Brain Signals	SQUID Sampling Rate
1kHz	10kHz

DAQ limited



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